

Commercial CO2 Sensor with MQTT Communication – Datasheet

1. Product Overview

WIN-SN-CO2-W is a versatile data acquisition device designed for real-time monitoring in industrial applications, especially ducts. It supports a wide range of external sensors with **UART interface**, providing flexibility for various measurements. Offering reliable communication and long-range data transfer over several meters, making it ideal for large-scale installations.

Its robust design makes it suitable for **installations in HVAC systems, environmental monitoring, and process control**. With **configurable data sampling rates, multi-sensor compatibility, and easy configuration**, WIN-SN-CO2-W ensures **accurate CO2 data acquisition and reliable performance** in demanding industrial environments, making it an excellent choice for **industrial and IOT applications**.



Fig.1 - Product Image

	Measuring Parameter	Measuring Range	Accuracy
☐	Carbon Dioxide Measurement Range	400 – 2000 PPM	±50.0 ppm ±5.0 %m.v.
☐	Carbon Dioxide Measurement Range	400 – 5000 PPM	±50.0 ppm ±5.0 %m.v.

Table 1

2. Sensor Specifications

- **Wide temperature range:** –40 °C to +125 °C
- **Full humidity range:** 0 % to 100 % RH
- **Power supply DC:** 12-24V@1A.
- **Product dimension:** 85*85*36 mm.
- **Certifications:**



3. Power Supply Connection

DC Connection

- Use 12-24V@ 1A, Power supply.
- Don't make loose connection.

4. Recommended Cable Electrical Characteristics

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

5. Device Configuration Process

1. Connect the Device

- Connect 12v-1A DC adaptor to power the board
- Connect the Wi-Fi SSID: WittelB-SN-CO2 Pass:123456789
- Enter the IP 192.168.4.1 in web browser
- Refer image 1 for Wi-Fi WittelB-SN-CO2

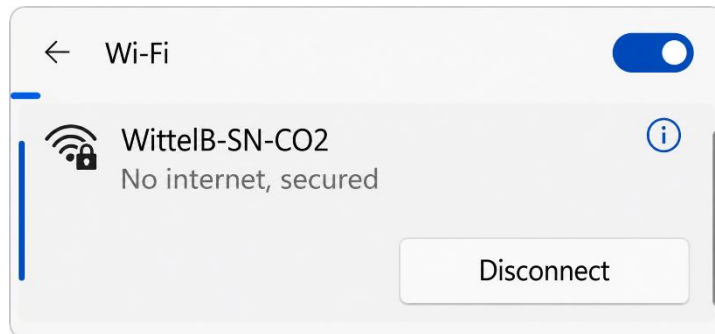
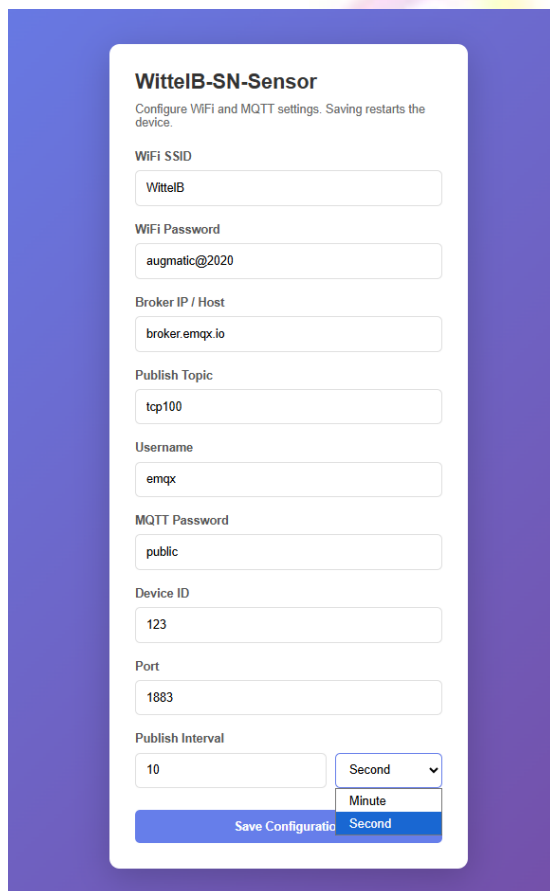


Image: 1



WittelB-SN-Sensor
Configure WiFi and MQTT settings. Saving restarts the device.

WiFi SSID
WittelB

WiFi Password
augmatic@2020

Broker IP / Host
broker.emqx.io

Publish Topic
tcp100

Username
emqx

MQTT Password
public

Device ID
123

Port
1883

Publish Interval
10 Second Minute Second

[Save Configuration](#)

Image: 2

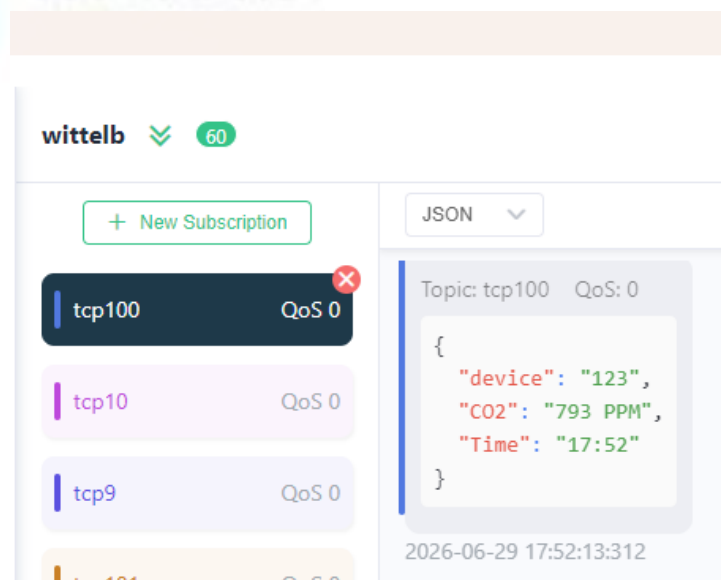


Image: 3

- Enter IP 192.168.4.1 in web browser then you will see the Web UI.
- Enter the local WiFi credentials as shown in above image2 so the device can connect to that WiFi network. Using this connection, it will send the sensor data to the cloud.
- Then enter the required MQTT Credentials and make sure the credentials should be correct.
- As shown in Image 2, you can select the publish interval in either minutes or seconds. MQTT data will be published according to the selected interval.
- Enter the MQTT credentials to transmit the sensor data to the cloud.
- Image 3 shows the sensor data at cloud for the selected topic.
- The published JSON payload includes a real-time timestamp, allowing each sensor reading to be tracked with its exact time.



Contact Information

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